A universal physics-informed neural network for used-LFP RUL: 50-cycle characterisation across three manufacturers

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Second-life repurposing of used lithium-iron-phosphate (LFP) cells for stationary energy storage is bottlenecked by characterisation cost. Cell-by-cell PyBaMM [1] rxn-limited SEI calibration reliably extrapolates for K400 cycles per cell [2]; below that the fitted slope becomes unstable. Prior second-life work [3, 4] improves calibration quality but does not shrink the cycling budget. *Can PyBaMM's rxn-limited SEI ODE be embedded as a physics prior inside a neural network, unlocking RUL prediction from far fewer measured cycles across LFP manufacturers?*

We train a joint PINN across all cells in a cohort: $SoH_\theta(n) = SoH_{init} - softplus(NN_\theta)$ with hard-monotonic decrement and per-cell latent embedding. Physics constraint (Ramadass-canonical rxn-limited SEI [5], grounded in PyBaMM's `Prada2013` LFP chemistry): $dSoH/dn = -k_{SEI}(i)\, SoH^{p(i)}$, with per-cell learnable (k_{SEI}, p) warm-started from a linear fit on the training window. A brief NN warm-up against the same linear-fade prior removes the flat-start artefact that degrades deep-fade cells at small K. Adam + cosine, 10k epochs, ≈ 9 min on a GPU.

Applied to 20 LFP cells across three manufacturers (7 MFR_A, 9 MFR_C, 4 MFR_B) with *identical* architecture and hyperparameters (K=50, ≈ 68k params), the PyBaMM-informed PINN achieves held-out RMSE < 4 pp SoH on **all 20 cells** (< 3 pp on 16/20, Fig. 1). On deep-second-life MFR_A cells (15–25 pp fade over 1000+ cycles), a linear extrapolation of the short-window SEI slope diverges (median 20 pp); the PINN maintains **max 3.7 pp — a 5$\times$ worst-case reduction** with 7/7 head-to-head wins (Fig. 2). On near-fresh MFR_B/MFR_C cells (1–3 pp fade) the linear form is already near-optimal (< 0.9 pp) and the PINN matches without harm. **Relative to per-cell PyBaMM calibration at K=400, the PINN reaches equivalent engineering accuracy at K=50 — 8$\times$ cycling reduction with make-agnostic transfer.**

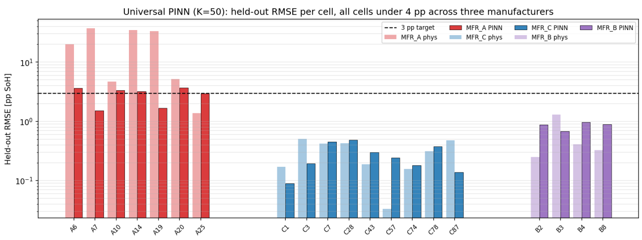

Figure 1: Per-cell held-out RMSE across 3 manufacturers (log scale). Short-window linear SEI-slope extrapolation towers above the 3 pp target on MFR_A; the PyBaMM-informed PINN sits at or below 3.7 pp on every cell.

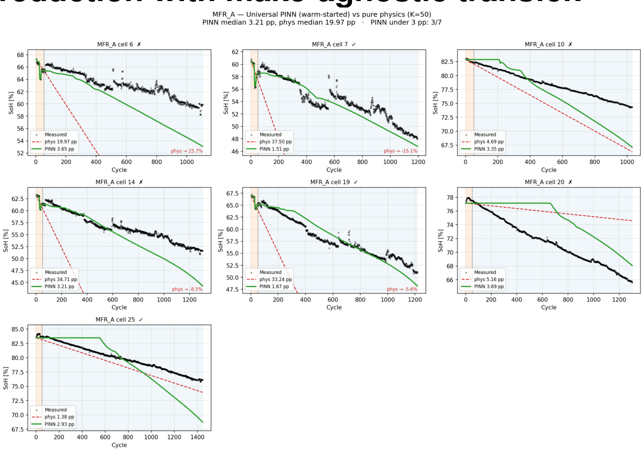

Figure 2: MFR_A trajectories (K=50). Naïve linear extrapolation exits the frame with impossible-SoH annotations; the PyBaMM-constrained PINN tracks measured across 1000+ cycles.

Results hold for LFP at 25 °C isothermal; cells flagged by an upstream shape filter (non-monotonic, three-regime, batch-transition data artefacts) remain out of scope. Reference K=400 calibration uses PyBaMM `v26.4.3`'s rxn-limited SEI implementation directly.

References. [1] V. Sulzer et al., *Python Battery Mathematical Modelling (PyBaMM)*, J. Open Res. Softw. 9(1), 14–21 (2021). [2] M. Prada et al., *Simplified electrochemical and thermal model of LiFePO$_4$-graphite Li-ion batteries*, J. Electrochem. Soc. 160(4), A616–A628 (2013). [3] O. Ouledboutaarija et al., *Advancing Second-Life Lithium-Ion Batteries*, PyBaMM Conf. 2025 Book of Abstracts, p. 19. [4] J. A. Kuzhiyil et al., *Enhancing Generalisability of Physics-based Battery Degradation Using UDE*, PyBaMM Conf. 2025 Book of Abstracts, p. 12. [5] P. Ramadass et al., *First-principles capacity fade model for Li-ion cells*, J. Electrochem. Soc. 151(2), A196–A203 (2004).